

# Journal Scope Drift Report

CWTS Leiden citation clustering · macro level · baseline 2020 → 2026 · run 20260721\_122750

|                                              |                                                 |                                                 |                                           |
|----------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------|
| <div>38,222</div> <div>Papers analysed</div> | <div>7.9%</div> <div>Overall out-of-scope</div> | <div>3 / 8</div> <div>High-drift journals</div> | <div>0.289</div> <div>Mean JSD 2026</div> |
|----------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------|

## Summary

This report looks at 8 Frontiers journals and asks a simple question: is the type of research each one publishes drifting away from where it started in 2020? Across all of them, about 1 in 13 papers (7.9%) now falls outside the journal's core subject areas. Three journals — Materials, Neurorobotics and Environmental Science — have moved the furthest from their original focus. A page of plain-language commentary and supporting detail follows for each journal.

## Scope Drift & OOS Summary by Journal

| Journal                   | Papers | OOS | OOS % | JSD 2026 | Drift    |
|---------------------------|--------|-----|-------|----------|----------|
| Fr. Neurorobotics         | 1,037  | 216 | 20.8% | 0.378    | High     |
| Fr. Robotics and AI       | 1,791  | 204 | 11.4% | 0.314    | Moderate |
| Fr. Chemistry             | 6,003  | 649 | 10.8% | 0.239    | Low      |
| Fr. Environmental Science | 7,093  | 660 | 9.3%  | 0.343    | High     |
| Fr. Materials             | 3,123  | 246 | 7.9%  | 0.379    | High     |
| Fr. Surgery               | 5,885  | 404 | 6.9%  | 0.232    | Low      |
| Fr. Aging Neuroscience    | 5,420  | 271 | 5.0%  | 0.145    | Low      |
| Fr. Earth Science         | 7,870  | 370 | 4.7%  | 0.280    | Moderate |

**A note on how to read this report:** This analysis is based purely on citation patterns (which papers cite which), not on reading the papers themselves. Citation-based methods like this can misclassify certain paper types — methods papers and meta-analyses in particular tend to cite (and be cited by) a much wider range of fields than their own subject matter would suggest, so they can appear more “out-of-scope” than they really are. Scope itself is also defined statistically here: a journal's “primary” clusters are simply whichever citation communities together account for 80% of its papers. This threshold is applied uniformly across all journals and hasn't been tuned per journal, so some journals may need a different cutoff to reflect their actual editorial scope. Treat the figures below as a useful signal for further investigation, not a definitive scope audit.

## Frontiers in Materials

High

Materials shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

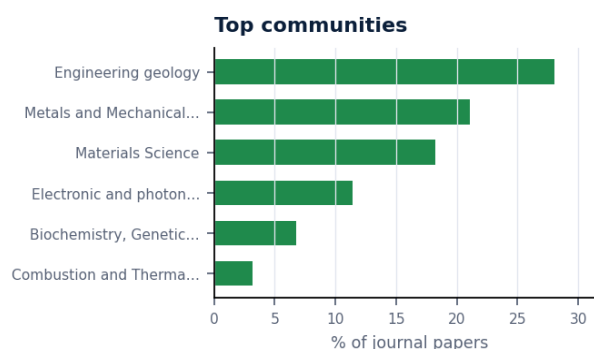
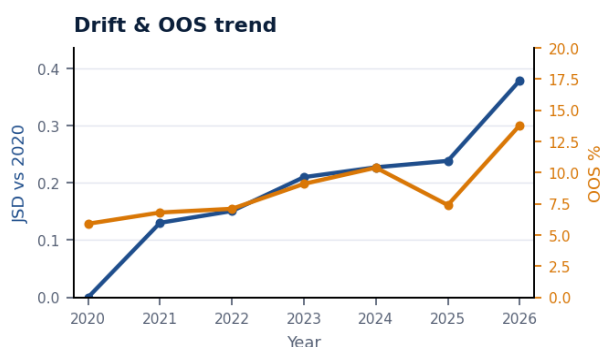
Right now, about 8% of Materials's papers (246 out of 3,123) fall outside its core subject clusters — mostly work related to Dentistry, Organic and Organometallic Chemistry. This out-of-scope share has risen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is "Engineering geology." Its primary clusters have changed since 2020. Areas that have left the core include Electronic and photonic properties. Newer core areas include Dentistry. In 2026 the largest core topics are Engineering geology, Metals and Mechanical Properties, and Materials Science (versus Materials Science, Metals and Mechanical Properties, and Engineering geology in 2020).

### Scope split

92% in-scope

8% OOS

|                        |                                   |                          |                                   |                              |                                  |
|------------------------|-----------------------------------|--------------------------|-----------------------------------|------------------------------|----------------------------------|
| <b>3,123</b><br>Papers | <b>246 (7.9%)</b><br>Out-of-scope | <b>0.379</b><br>JSD 2026 | <b>-0.228</b><br>Entropy $\Delta$ | <b>5</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|-----------------------------------|--------------------------|-----------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                                    | Status   | Papers | Share of journal |
|----------------------------------------------|----------|--------|------------------|
| Engineering geology                          | In-scope | 875    | 28.0%            |
| Metals and Mechanical Properties             | In-scope | 660    | 21.1%            |
| Materials Science                            | In-scope | 567    | 18.2%            |
| Electronic and photonic properties           | In-scope | 355    | 11.4%            |
| Biochemistry, Genetics and Molecular Biology | In-scope | 213    | 6.8%             |
| Combustion and Thermal Management            | In-scope | 100    | 3.2%             |

### Example papers

#### In-scope

- An in-depth study on tribological behaviour of polymers and polymer composites: state-of-the-art
- Investigation of rutting deformation mechanisms and high-temperature deformation resistance of asphalt mixtures
- Research on factors influencing high temperature deformation of asphalt mixture and prediction modeling
- Effect of fly ash and metakaolin on early-age autogenous shrinkage and mechanical properties of ultra-high-performance concrete
- Mechanically stimulated self-powered electrochemical sensors: principles, classifications, and future directions

#### Out-of-scope

- Progress and biomedical applications of antimicrobial peptide-functionalized titanium surfaces
- Enhancement of zirconia–veneering porcelain bond strength via thermal phosphoric acid etching combined with lithium Silicate Coating and the influence of multiple sintering cycles
- Fungal-derived functional carbons for secondary batteries: from biomass architectures to electrochemical performance
- A strategy for fabricating a multifunctional hydrogel with enhanced adhesion and ROS-scavenging capabilities
- Engineering D-CONGA-Q7–embedded hydrogel matrices: physicochemical characterization and antimicrobial functionality

## Frontiers in Neurorobotics

**High**

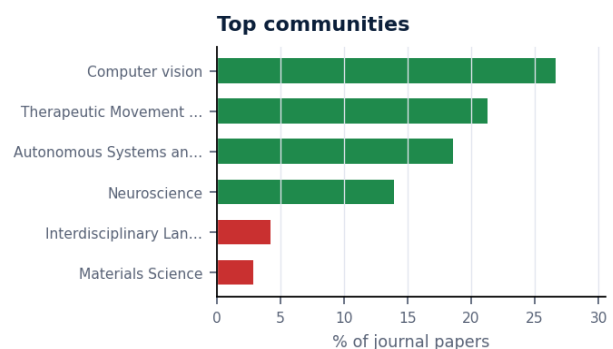
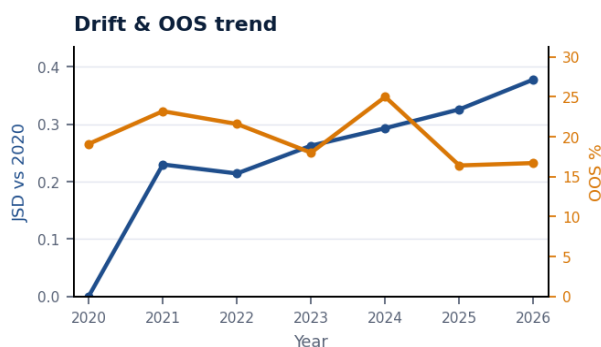
Neurorobotics shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

Right now, about 21% of Neurorobotics's papers (216 out of 1,037) fall outside its core subject clusters — mostly work related to Interdisciplinary Language and Health Technology, Materials Science. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Computer vision.” Its primary clusters have not meaningfully changed since 2020 — still centred on Therapeutic Movement Sciences, Neuroscience, and Autonomous Systems and Control Engineering.

### Scope split

**79% in-scope**
**21% OOS**

|                        |                                    |                          |                                    |                              |                                  |
|------------------------|------------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|
| <b>1,037</b><br>Papers | <b>216 (20.8%)</b><br>Out-of-scope | <b>0.378</b><br>JSD 2026 | <b>-0.4157</b><br>Entropy $\Delta$ | <b>4</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|------------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                                        | Status       | Papers | Share of journal |
|--------------------------------------------------|--------------|--------|------------------|
| Computer vision                                  | In-scope     | 276    | 26.6%            |
| Therapeutic Movement Sciences                    | In-scope     | 221    | 21.3%            |
| Autonomous Systems and Control Engineering       | In-scope     | 193    | 18.6%            |
| Neuroscience                                     | In-scope     | 144    | 13.9%            |
| Interdisciplinary Language and Health Technology | Out-of-scope | 44     | 4.2%             |
| Materials Science                                | Out-of-scope | 30     | 2.9%             |

### Example papers

#### In-scope

- Research on embodied agent multimodal perception and real-time path planning algorithms for complex unstructured environments
- Transformer-based human-motion forecasting coupled with safe reinforcement learning for telepresence robot co-navigation
- Emotion estimation from video footage with LSTM
- Interpretable side-aware kinematic-sEMG gait-state representations relevant to adaptive neurorobotic assistance after stroke: a public-dataset study
- Interpretable vision mamba for SAR images classification

#### Out-of-scope

- Motion feature extraction based on semi-supervised learning and long short-term memory network in digital dance
- Design an anthropomorphic dexterous hand for expressive piano performance
- A lightweight Chinese–English translation model integrating compressed BERT attention and phrase discard mechanism
- Human-inspired sensorimotor controller for dynamic motion adaptation: a study in robotic arms
- Innovative approach of nonlinear controllers design for prosthetic knee performance

## Frontiers in Environmental Science

High

Environmental Science shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.34) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

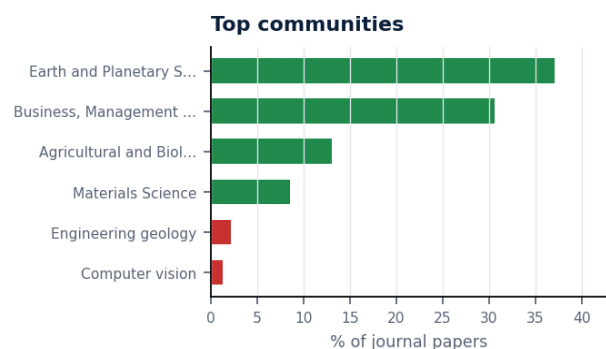
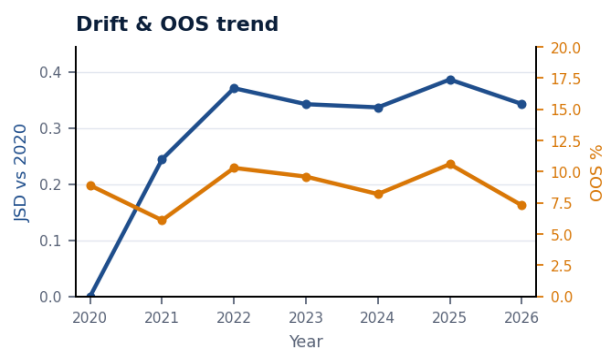
Right now, about 9% of Environmental Science's papers (660 out of 7,093) fall outside its core subject clusters — mostly work related to Engineering geology, Computer vision. This out-of-scope share has fallen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is “Earth and Planetary Sciences.” Its primary clusters have changed since 2020. Areas that have left the core include Materials Science. Newer core areas include Business, Management and Accounting. In 2026 the largest core topics are Earth and Planetary Sciences, Business, Management and Accounting, and Agricultural and Biological Sciences (versus Earth and Planetary Sciences, Materials Science, and Agricultural and Biological Sciences in 2020).

### Scope split

91% in-scope

9% OOS

|                        |                                   |                          |                                    |                              |                                  |
|------------------------|-----------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|
| <b>7,093</b><br>Papers | <b>660 (9.3%)</b><br>Out-of-scope | <b>0.343</b><br>JSD 2026 | <b>-0.0436</b><br>Entropy $\Delta$ | <b>3</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|-----------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                            | Status       | Papers | Share of journal |
|--------------------------------------|--------------|--------|------------------|
| Earth and Planetary Sciences         | In-scope     | 2,629  | 37.1%            |
| Business, Management and Accounting  | In-scope     | 2,171  | 30.6%            |
| Agricultural and Biological Sciences | In-scope     | 926    | 13.1%            |
| Materials Science                    | In-scope     | 608    | 8.6%             |
| Engineering geology                  | Out-of-scope | 156    | 2.2%             |
| Computer vision                      | Out-of-scope | 91     | 1.3%             |

### Example papers

#### In-scope

- Spatial distribution characteristics and optimization strategies of carbon sequestration service in the yellow river Basin ecosystems: a case study of Shanxi section
- A 20-year spatio-temporal analysis of 3D radar-based hail tracks in Germany: trends and regional differences
- A novel approach to TDS estimation in desalination systems using differential pressure sensing

#### Out-of-scope

- An interpretable building air conditioning load forecasting framework using SSA-optimized Bi-LSTM and SHAP analysis
- Does digital finance drive the green transformation of the industrial sector? Evidence from the panel data of Chinese cities
- Bridging the green and gold divide: exploring the impact of green finance policy and digital finance on ecological product value realization

## Frontiers in Robotics and AI

Moderate

Robotics and AI is showing a moderate amount of movement away from its 2020 starting point (JSD 0.31). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

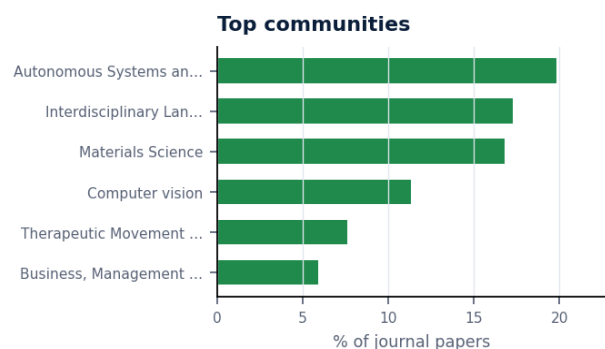
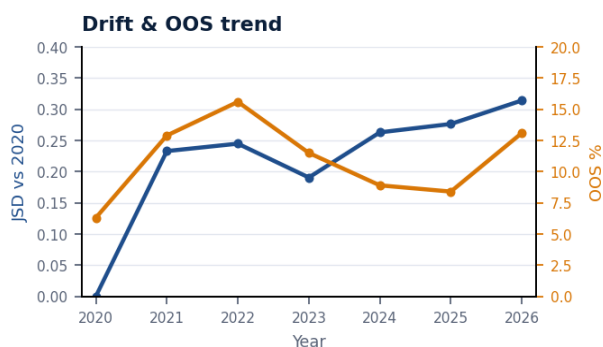
Right now, about 11% of Robotics and AI's papers (204 out of 1,791) fall outside its core subject clusters — mostly work related to Space physics, Integrated Telecommunications and Electromagnetics. This out-of-scope share has risen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is "Autonomous Systems and Control Engineering." Its primary clusters have changed since 2020. Newer core areas include Business, Management and Accounting. In 2026 the largest core topics are Autonomous Systems and Control Engineering, Interdisciplinary Language and Health Technology, and Computer vision (versus Autonomous Systems and Control Engineering, Materials Science, and Therapeutic Movement Sciences in 2020).

### Scope split

89% in-scope

11% OOS

|                        |                                    |                          |                                    |                              |                                  |
|------------------------|------------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|
| <b>1,791</b><br>Papers | <b>204 (11.4%)</b><br>Out-of-scope | <b>0.314</b><br>JSD 2026 | <b>+0.0379</b><br>Entropy $\Delta$ | <b>7</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|------------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                                        | Status   | Papers | Share of journal |
|--------------------------------------------------|----------|--------|------------------|
| Autonomous Systems and Control Engineering       | In-scope | 354    | 19.8%            |
| Interdisciplinary Language and Health Technology | In-scope | 309    | 17.3%            |
| Materials Science                                | In-scope | 301    | 16.8%            |
| Computer vision                                  | In-scope | 203    | 11.3%            |
| Therapeutic Movement Sciences                    | In-scope | 136    | 7.6%             |
| Business, Management and Accounting              | In-scope | 106    | 5.9%             |

### Example papers

#### In-scope

- Social robots: a meta-analysis of learning outcomes
- Modelling C-arm fluoroscopy and operating table kinematics via machine learning
- SANE (eaSy gAit aNalysis systEm): real-time dual-depth, marker-less system characterisation and robustness of gait parameters
- Multi-strategy Sea Horse Optimization algorithm for UAV path planning
- RAISE-FER: a massive cross-dataset augmented facial expression dataset

#### Out-of-scope

- Bridging ancient wisdom and cognitive engagement: a comparative study of chatbot-based moral instruction
- Data-driven analysis of Armeo Spring performance across neurological disorders: implications for personalized upper limb neurorehabilitation
- Human-AI co-research on design and evaluation of Embodied Conversational Agent in rehabilitation contexts
- Optimizing avian flight dynamics with a synergetic bio-inspired and machine learning approach
- Accuracy of robot-assisted anterior transpedicular screws in the subaxial cervical spine: an experimental study on human specimens

## Frontiers in Earth Science

Moderate

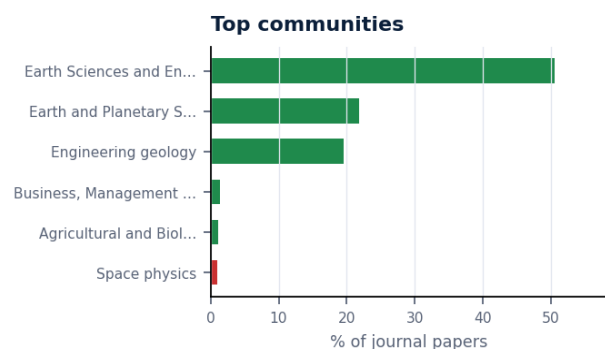
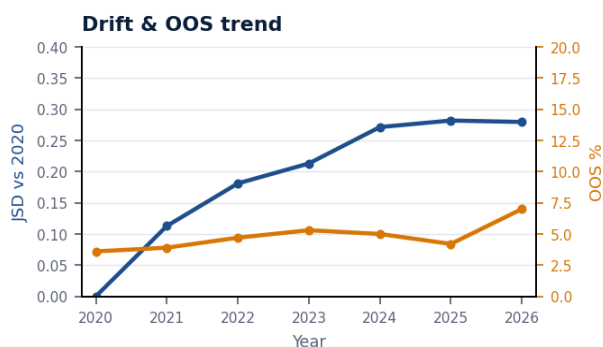
Earth Science is showing a moderate amount of movement away from its 2020 starting point (JSD 0.28). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

Right now, about 5% of Earth Science's papers (370 out of 7,870) fall outside its core subject clusters — mostly work related to Space physics, Computer vision. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is "Earth Sciences and Energy Resources." Its primary clusters have changed since 2020. Newer core areas include Engineering geology. In 2026 the largest core topics are Earth Sciences and Energy Resources, Engineering geology, and Earth and Planetary Sciences (versus Earth Sciences and Energy Resources and Earth and Planetary Sciences in 2020).

### Scope split

95% in-scope

|                        |                                   |                          |                                    |                              |                                  |
|------------------------|-----------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|
| <b>7,870</b><br>Papers | <b>370 (4.7%)</b><br>Out-of-scope | <b>0.280</b><br>JSD 2026 | <b>+0.1262</b><br>Entropy $\Delta$ | <b>3</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|-----------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                            | Status       | Papers | Share of journal |
|--------------------------------------|--------------|--------|------------------|
| Earth Sciences and Energy Resources  | In-scope     | 3,981  | 50.6%            |
| Earth and Planetary Sciences         | In-scope     | 1,716  | 21.8%            |
| Engineering geology                  | In-scope     | 1,534  | 19.5%            |
| Business, Management and Accounting  | In-scope     | 109    | 1.4%             |
| Agricultural and Biological Sciences | In-scope     | 87     | 1.1%             |
| Space physics                        | Out-of-scope | 80     | 1.0%             |

### Example papers

#### In-scope

- A novel sedimentologically controlled accumulation pattern for marine shale gas: insights from the "source-reservoir-seal" configurations of lower Carboniferous shale in the Yaziluo rift trough, Southern China
- Machine learning-based flood inundation mapping using fused optical and SAR remote sensing: a case study of the Miyun flood
- Provenance and tectonic setting of the Lower Jurassic-Lower Cretaceous shales from the northeastern Iraq: mineralogical and geochemical approaches
- Multiscale characterization of pore-throat structures and flow mechanisms in low-permeability reservoirs
- Mapping the impacts of recurrent floods and windstorms in Bauchi, Nigeria: a hybrid multi-criteria approach using fuzzy-AHP

#### Out-of-scope

- Prediction of the compressive strength of tailings backfill using an EO-LightGBM model: performance comparison and feature importance analysis
- Study on the dynamic response characteristics of hydraulic support columns under multi-condition impact loads
- Multi scale attention driven GIS framework for geoinformatics teaching applications in technical colleges
- Semi-automated bed boundary detection and log alignment using fast Walsh-Hadamard transform
- MVarGOSIM: an MPS algorithm for characterizing complex structures with multiple variables

## Frontiers in Chemistry

Low

Chemistry remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.24), which suggests the journal is publishing consistently within its intended focus.

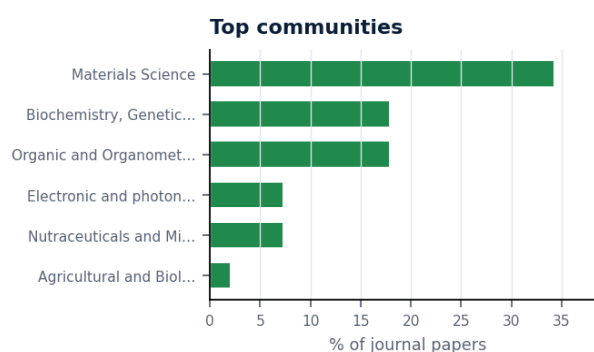
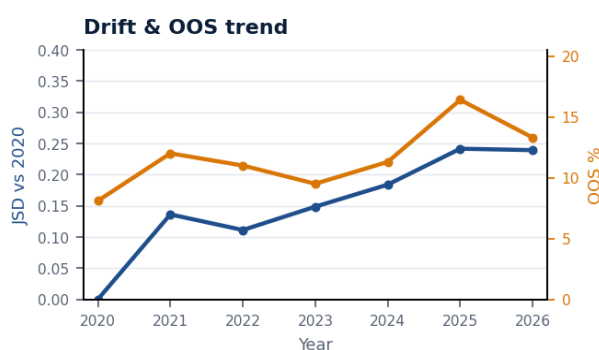
Right now, about 11% of Chemistry's papers (649 out of 6,003) fall outside its core subject clusters — mostly work related to Infectious Disease and Public Health, Pathogenesis and Resistance. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is "Materials Science." Its primary clusters have changed since 2020. Areas that have left the core include Electronic and photonic properties. Newer core areas include Nutraceuticals and Microbiome Interactions and Intelligent Decision-Making Systems. In 2026 the largest core topics are Materials Science, Biochemistry, Genetics and Molecular Biology, and Organic and Organometallic Chemistry (versus Materials Science, Organic and Organometallic Chemistry, and Biochemistry, Genetics and Molecular Biology in 2020).

### Scope split

89% in-scope

11% OOS

|                        |                                    |                          |                                    |                              |                                  |
|------------------------|------------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|
| <b>6,003</b><br>Papers | <b>649 (10.8%)</b><br>Out-of-scope | <b>0.239</b><br>JSD 2026 | <b>+0.2844</b><br>Entropy $\Delta$ | <b>5</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|------------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                                    | Status   | Papers | Share of journal |
|----------------------------------------------|----------|--------|------------------|
| Materials Science                            | In-scope | 2,051  | 34.2%            |
| Biochemistry, Genetics and Molecular Biology | In-scope | 1,068  | 17.8%            |
| Organic and Organometallic Chemistry         | In-scope | 1,067  | 17.8%            |
| Electronic and photonic properties           | In-scope | 434    | 7.2%             |
| Nutraceuticals and Microbiome Interactions   | In-scope | 430    | 7.2%             |
| Agricultural and Biological Sciences         | In-scope | 123    | 2.0%             |

### Example papers

#### In-scope

- HIT101308137 and HIT104293658 nominate dual target chemotypes for PTPN1 and PTPN2 with preliminary selectivity in colorectal cancer cells
- Editorial: Exploration of the role of heme proteins in biology with experimental and computational methods
- Formation and transfer patterns of key aroma compounds in cigar tobacco based on SBSE-GC-O-MS, pyrolysis-GC-O-MS and flue gas analysis correlation studies
- Diol-enhanced natural deep eutectic solvents for efficient poplar pretreatment
- Influence of substitution patterns on the antimicrobial properties of pyrrole sulfonamide scaffolds

#### Out-of-scope

- Predictive modeling and ranking of physicochemicals in Aegle marmelos using topological indices and multi-criteria decision-making techniques
- Face reverse degree topological analysis of TP-COFs, existence of isentropic COFs and spectral characteristics
- Entropy comparison of benzothiadiazole-based covalent-organic frameworks
- Trypanosomatid parasites, translational research in a One Health, One World environment
- Membrane-targeted schiff base derivatives overcome MRSA resistance through phosphatidylglycerol binding and ROS-mediated killing



## Frontiers in Surgery

Low

Surgery remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.23), which suggests the journal is publishing consistently within its intended focus.

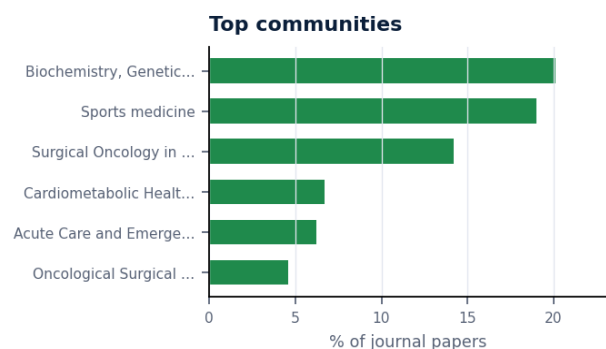
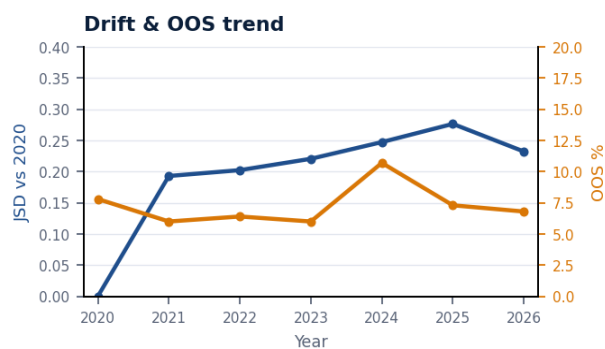
Right now, about 7% of Surgery's papers (404 out of 5,885) fall outside its core subject clusters — mostly work related to peripheral topics. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Biochemistry, Genetics and Molecular Biology.” Its primary clusters have changed since 2020. Areas that have left the core include Infectious Disease and Public Health and Audiology and Hearing. Newer core areas include Neuroscience and Endocrine-Oncology Surgical Pathology. In 2026 the largest core topics are Sports medicine, Biochemistry, Genetics and Molecular Biology, and Surgical Oncology in Hepatic Disease (versus Biochemistry, Genetics and Molecular Biology, Surgical Oncology in Hepatic Disease, and Sports medicine in 2020).

### Scope split

93% in-scope

7% OOS

|                        |                                   |                          |                             |                               |                                  |
|------------------------|-----------------------------------|--------------------------|-----------------------------|-------------------------------|----------------------------------|
| <b>5,885</b><br>Papers | <b>404 (6.9%)</b><br>Out-of-scope | <b>0.232</b><br>JSD 2026 | <b>-0.1047</b><br>Entropy Δ | <b>10</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|-----------------------------------|--------------------------|-----------------------------|-------------------------------|----------------------------------|



### Community composition

| Community                                    | Status   | Papers | Share of journal |
|----------------------------------------------|----------|--------|------------------|
| Biochemistry, Genetics and Molecular Biology | In-scope | 1,181  | 20.1%            |
| Sports medicine                              | In-scope | 1,119  | 19.0%            |
| Surgical Oncology in Hepatic Disease         | In-scope | 834    | 14.2%            |
| Cardiometabolic Health Sciences              | In-scope | 397    | 6.7%             |
| Acute Care and Emergency Medicine            | In-scope | 362    | 6.2%             |
| Oncological Surgical Pathology Research      | In-scope | 272    | 4.6%             |

### Example papers

#### In-scope

- Establishing an implementation framework for clinical nursing guidelines in hepatobiliary and accelerated rehabilitation surgery: based on the Ottawa research application model
- Case Report: Arthroscopic synovectomy and loose body removal for charcot knee in an adolescent with congenital insensitivity to pain with anhidrosis
- Comparative outcomes of natural orifice specimen extraction surgery versus totally laparoscopic surgery for right-sided colon cancer: a single-centre propensity score-matched study

#### Out-of-scope

- A refined radiological classification of anterior clinoid process pneumatization
- Editorial: Microflora and bacterial translocation in intestinal obstruction
- Effects of rhythmic dumbbell upper limb rehabilitation training based on the multi-process action control framework on vascular access function and quality of life in hemodialysis patients



## Frontiers in Aging Neuroscience

Low

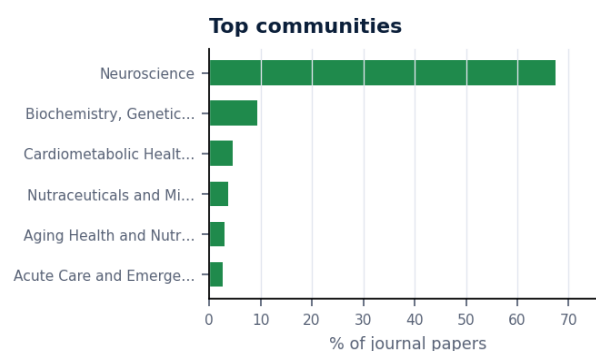
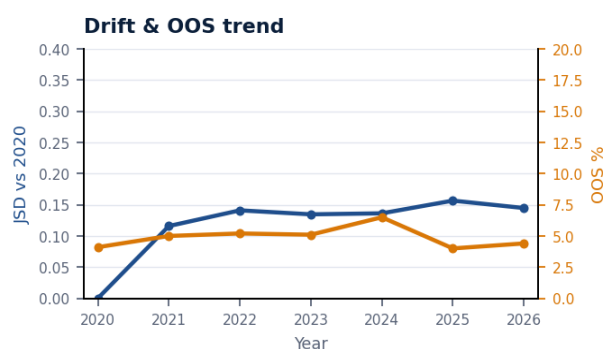
Aging Neuroscience remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.14), which suggests the journal is publishing consistently within its intended focus.

Right now, about 5% of Aging Neuroscience's papers (271 out of 5,420) fall outside its core subject clusters — mostly work related to Ophthalmology, Pediatric Neurological Critical Care. This out-of-scope share has stayed similar since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is "Neuroscience." Its primary clusters have changed since 2020. Newer core areas include Cardiometabolic Health Sciences. In 2026 the largest core topics are Neuroscience, Biochemistry, Genetics and Molecular Biology, and Cardiometabolic Health Sciences (versus Neuroscience and Biochemistry, Genetics and Molecular Biology in 2020).

### Scope split

95% in-scope

|                        |                                   |                          |                                    |                              |                                  |
|------------------------|-----------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|
| <b>5,420</b><br>Papers | <b>271 (5.0%)</b><br>Out-of-scope | <b>0.145</b><br>JSD 2026 | <b>+0.1983</b><br>Entropy $\Delta$ | <b>3</b><br>Primary clusters | <b>80.0%</b><br>Primary coverage |
|------------------------|-----------------------------------|--------------------------|------------------------------------|------------------------------|----------------------------------|



### Community composition

| Community                                    | Status   | Papers | Share of journal |
|----------------------------------------------|----------|--------|------------------|
| Neuroscience                                 | In-scope | 3,652  | 67.4%            |
| Biochemistry, Genetics and Molecular Biology | In-scope | 503    | 9.3%             |
| Cardiometabolic Health Sciences              | In-scope | 244    | 4.5%             |
| Nutraceuticals and Microbiome Interactions   | In-scope | 195    | 3.6%             |
| Aging Health and Nutrition Research          | In-scope | 163    | 3.0%             |
| Acute Care and Emergency Medicine            | In-scope | 143    | 2.6%             |

### Example papers

#### In-scope

- Identification and validation of an interpretable EEG-based machine learning model for the diagnosis of post-stroke cognitive impairment
- Neural and motor mechanisms of handwriting: from healthy aging to neurodegenerative disorders
- Neuromelanin, iron and MRI measurements in midbrain tissues of Parkinson's and Alzheimer's subjects
- Age-dependent vulnerability to spatial memory interference in APP/PS1 mice
- Mechanisms to medicines: navigating drug repurposing strategies in Alzheimer's disease

#### Out-of-scope

- Propranolol reinstates mitochondrial dynamics and synaptic memory pathways through CaMKII/CREB-BDNF/ PKM $\zeta$  cascades in an AD-like rat model
- Alterations in energy production in a Drosophila model for the X-linked dystonia-parkinsonism-related Taf1 deficiency
- Systemic delivery of anti-sense oligonucleotide targeting  $\alpha$ -synuclein for treatment in a mouse model of multiple system atrophy
- Needle-warming moxibustion alleviates pain in rats with cervical spondylotic radiculopathy by modulating the NF- $\kappa$ B/ROS/NLRP3 pathway
- Protective effect of baicalein from Pinellia ternate on Alzheimer's disease cell injury: a network pharmacology, molecular docking, and molecular dynamics study